

Nicholas Woolsey

EDUCATION

Bachelor of Science: Mechanical Engineering Major — 3.37 GPA —
Electrical Engineering Minor

August 2023 - May 2026, University of Colorado - Boulder

Front Range Community College — 3.35 GPA —

August 2020 - December 2023

- Member of Phi Theta Kappa Honor Society.
- Received Dean's List award in last 4 semesters.

ENGINEERING POSITIONS/INTERNSHIPS

MDS II Student Researcher

May 2024 - May 2026, University of Colorado, Boulder, CO

Dr. Jacob Segil

- Conducted medical device research, improving the design of a femoral nail, and began research on Total Hip Replacement.
- Conducted stress testing on nail samples.
- Primary designer on Sciatic Notch Retractor devices, emphasizing design-for-manufacturability and working with machinists to create complex part drawings.
- Primary operator of the Formlabs Fuse 1 SLS 3D printer.
- Began a project on a digitalized femoral nail goniometer.

Robotics Workshop Leader Intern

May 2023 - August 2023, Colorado Space Grant Consortium, Boulder, CO

- Mentored 30+ college students in the fundamentals of Arduino, soldering, coding, and robotics, accumulating valuable leadership and mentor experience.
- Designed, tested, and facilitated workshops and delivered presentations.

National Science Foundation Research Experience for Undergrads

May 2022 - August 2022, Colorado State University, Fort Collins, CO

Dr. Jianguo Zhao

- Researched a modified passive bistable gripper for drone perching.
- Implemented spring steels into multiple original designs modeled in Fusion 360, utilizing FDA/SLA 3D printing and Arduino-based robotics.
- Presented results using strong verbal and written communication skills to professors and graduate students.

Contact

Location

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LinkedIn

www.linkedin.com/in/nwoolsey

Portfolio

nickw36.github.io/NicholasWoolsey

Relevant Coursework

Thermodynamics, Heat Transfer, Materials Science, Linear Systems, Mechatronics 1&2, Fluid Dynamics, Component Design, Senior Design, Manufacturing Processes & Systems

Skills

Research and Analysis



Research Design



Proficient in C++



CAD/CAM expertise



Lathes and mills manufacturing



Certifications

2020-05 - SolidWorks Additive Manufacturing Associate

2021-04 - SolidWorks Mechanical Design Associate

Engineer Intern (EI) | State of Colorado | License #EI.0081447

2025-12 - Fundamentals of Engineering (FE) Exam

Interests

Calisthenics

Christian

ENGINEERING EXPERIENCES

CU Robotics Manufacturing Lead

September 2023 - May 2026

- Lead the CAM process by collaborating with design engineers, transferring CAD models to CAM operations in Fusion 360 to create and refine toolpaths for CNC machining.
- Perform hands-on CNC manufacturing of robotic components using manual lathes and 3-axis mills, gaining direct machine operation and setup exposure on the shop floor.
- Apply knowledge of machining strategy, workholding, and tooling selection to ensure manufacturability and maintain dimensional tolerances.
- Co-lead meetings, as well as train and prepare material for incoming manufacturing members.

Senior Design: Micro Motion Drop Tester

August 2025 - May 2026

- Lead Electromechanical Integration: Designed the interface between control software and hardware, utilizing Arduino and serial communication for remote hoist actuation.
- Designed and prototyped application-specific PCBs to centralize signal processing and streamline wiring.
- Engineered a 120V AC safety subsystem featuring a 10A chain hoist, hardware interlocks, and E-stop integration to ensure fail-safe operation.
- People's Choice, 2nd out of 169 projects

NASA L-SPACE Mission Concept Academy

January 2024 - April 2024

- Gained greater insight and skills in NASA mission protocols, procedures
- Dedicated 10-15 hours weekly to rigorous, deep technical training in mission planning and proposal writing.
- Collaborated with an interdisciplinary team to develop a NASA mission concept, from defining science objectives, designing a spacecraft for the mission, and presenting a preliminary design review.
- Received all skill badges, including: Teaming, Requirements, Risk Management, Project Management, Systems, Heat Transfer, and Siemens NX CAD.

NASA Community College Aerospace Scholars Alumni - 3 Missions

July 2023 - February 2024

- Delivered infographics for a simulated Mars mission for missions 1 & 2.
- 1 of 230 selected for Mission 3 to continue the program at a NASA facility.
- Presented an Urban Air Mobility proposal for eVTOL vertiports to NASA engineers at Langley Research Center.
- Developed proof of concept in Fusion 360, demonstrating unique and modular battery swapping, eVTOL storage facility, and hub.

NASA COSGC Robotics Challenge

September 2021 - April 2022

- Built an autonomous obstacle avoidance rover in a team of 2, competing alongside other state colleges at the Great Sand Dunes.
- Acquired experience in circuits, C++, soldering, CAD, and 3D printing.
- Utilized Fusion 360 to create an optimized rover chassis using 50% less material while maintaining structural strength, resulting in the Best Structure Award.
- Completed all 5 Mars simulation courses at the challenge event and received a Certificate of Achievement and top honors for robotics poster.